

ETH ZÜRICH 5-YEAR PREPARATION GUIDE AND QUESTION BANK

PART 4: Final Mock Exams, Exam Strategies & 40 Questions — The Final Level

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Series Book 4 — The Final Chapter

"The journey ends here. The exam begins now."

This is the Final Part of the Series

This file is the **4th and final part** of the **ETH Zürich 5-Year Preparation Guide and Question Bank** series. The complete series:

- **Part 1:** General tactics, study plan, 5-year roadmap, mathematics and foundational physics.
- **Part 2:** Physics (electricity, magnetism, thermodynamics, waves), biology, chemistry, German language.
- **Part 3:** Advanced mathematics, advanced physics, German C1 academic reading/writing.
- **Part 4 (this book):** Final mock exams, exam strategies, time management, and 40 comprehensive exam-style questions across all subjects.

Recommendation: Complete Parts 1–3 before starting this final part.

Final Advice: The ETH Zürich entrance exam is about calm, structured, and disciplined thinking under pressure. Trust the preparation you have done.

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This section is yours. Click any cell and type.

Day	Focus	Task	Time
Monday			60 min
Tuesday			60 min
Wednesday			60 min
Thursday			60 min
Friday			60 min
Saturday	Mock Exam	Full timed simulation	3 h
Sunday	Rest	Light review only	—

Item	Done?
All Part 1–4 questions solved at least once	☐
Error Log reviewed — no repeated mistakes	☐
Full mock exam done under timed conditions	☐
German C1 writing practiced (Erörterung + Zusammenfassung)	☐
Formula sheet memorized (math + physics)	☐
Sleep schedule adjusted to exam days	☐
Documents & ID prepared (if applicable)	☐
Mental preparation: calm, focused, ready	☐

Section 2: Exam Strategies & Time Management

1. Before the Exam

- **Sleep:** At least 7–8 hours the night before. No all-nighters.
- **Nutrition:** Light breakfast, water bottle. Avoid heavy food.
- **Materials:** Pencils, eraser, ruler, calculator (if allowed), ID.
- **Arrive early:** 20–30 minutes before, to settle in and avoid panic.

2. During the Exam — Time Management

Phase	Time	Strategy
Read-through	5 min	Scan all questions. Mark easy / medium / hard.
Easy questions first	30% of time	Secure guaranteed points. Build confidence.
Medium questions	40% of time	Main bulk of the exam.
Hard questions	20% of time	Attempt only if easy/medium are done.
Review	10% of time	Check units, signs, arithmetic, skipped parts.

3. If You Get Stuck

- **Move on:** Don't waste more than 5 minutes on one question.
- **Write what you know:** Partial credit is real. Write the formula, approach, first step.
- **Draw a diagram:** Even a rough sketch can unlock the solution.
- **Check units:** Often a dimensional analysis reveals the mistake.

4. Oral Exam (Mündliche Prüfung)

- **Narrate your thinking:** The examiner wants to hear your reasoning.
- **Speak in complete sentences (German/English):** "First, I will resolve the forces into their components..."
- **Self-correct aloud:** "Wait, my unit check fails here — I must have dropped a factor."

5. After the Exam

- Do NOT compare answers with others immediately.
- Write down what felt difficult while it's fresh.
- Rest. The next subject needs your energy too.

Section 3: Mock Exam 1 (Q1–Q10)

Time yourself. No notes. No calculator unless allowed.

Q1 Medium Calculus 6m

EN: Find $\frac{dy}{dx}$ if $y = \ln(\cos(x)) + e^{3x} \sin(x)$.

DE: Bestimmen Sie $\frac{dy}{dx}$ für $y = \ln(\cos(x)) + e^{3x} \sin(x)$.

💡 **Strategy:** Chain rule + product rule.

Solution: $-\tan(x) + e^{3x}(3\sin(x) + \cos(x))$.

🔥 Working space

Q2 Hard Integration 8m

EN: Evaluate $\int_0^1 \frac{2x}{x^2+1} dx$.

DE: Berechnen Sie $\int_0^1 \frac{2x}{x^2+1} dx$.

💡 **Strategy:** Substitution $u = x^2 + 1$.

Solution: $\ln 2$.

🔥 Working space

Q3 Hard Vectors 8m

EN: Given $\vec{a} = (2, -1, 3)$ and $\vec{b} = (1, 4, -2)$, find $\vec{a} \times \vec{b}$ and the angle between them.

DE: Für $\vec{a} = (2, -1, 3)$ und $\vec{b} = (1, 4, -2)$: Kreuzprodukt und Winkel.

Solution: $\vec{a} \times \vec{b} = (-10, 7, 9)$; $\cos \theta = \frac{-8}{\sqrt{14}\sqrt{21}}$.

🔥 Working space

Q4 Medium Kinematics 8m

EN: A ball is thrown upward with initial velocity 15 m/s. Find the maximum height and total time in the air ($g = 10 \text{ m/s}^2$).

DE: Ein Ball wird mit 15 m/s nach oben geworfen. Maximale Höhe und Gesamtflugzeit ($g = 10 \text{ m/s}^2$).

Solution: $h_{\max} = 11.25 \text{ m}$, $t = 3 \text{ s}$.

 Working space

Q5 Hard Energy 8m

EN: A 3 kg block slides down a rough incline ($\theta = 45^\circ$, $\mu_k = 0.3$) from a height of 2 m. Find its speed at the bottom.

DE: Ein 3 kg Block rutscht eine raue schiefe Ebene hinab ($\theta = 45^\circ$, $\mu_k = 0.3$) aus 2 m Höhe. Geschwindigkeit unten?

Solution: Use energy: $mgh - \mu mg \cos \theta \cdot d = \frac{1}{2}mv^2$. With $d = h / \sin \theta$, $v \approx 4.8 \text{ m/s}$.

 Working space

Q6 Medium Circuits 6m

EN: Two resistors $R_1 = 6 \Omega$ and $R_2 = 3 \Omega$ are in parallel, connected to a 12 V battery. Find total current and power on R_1 .

DE: $R_1 = 6 \Omega$, $R_2 = 3 \Omega$ parallel, 12 V. Gesamtstrom und Leistung an R_1 .

Solution: $R_{eq} = 2 \Omega$, $I = 6 \text{ A}$, $P_1 = 24 \text{ W}$.

 Working space

Q7 **Hard** **Optimization** **10m**

EN: A box with a square base (side x) and no lid has volume 32 m^3 . Find the dimensions that minimize the surface area.

DE: Eine offene Kiste mit quadratischer Grundfläche (x) hat Volumen 32 m^3 . Minimieren Sie die Oberfläche.

Solution: $x = 4 \text{ m}$, height $h = 2 \text{ m}$. $SA_{min} = 48 \text{ m}^2$.

 Working space

Q8 **Hard** **Magnetic** **8m**

EN: A wire of length 0.5 m carrying 4 A is placed perpendicular to a magnetic field of 0.3 T . Find the force on the wire.

DE: Ein 0.5 m langer Draht mit 4 A senkrecht zu $B = 0.3 \text{ T}$. Kraft?

Solution: $F = BIL = 0.6 \text{ N}$.

 Working space

Q9 **Medium** **Complex** **6m**

EN: Compute $(1 + i)^4$ and express the result in $a + bi$ form.

DE: Berechnen Sie $(1 + i)^4$ in der Form $a + bi$.

Solution: $= -4$.

 Working space

Q10 Hard Series 8m

EN: Determine whether $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges. If so, find its sum.

DE: Konvergiert $\sum_{n=1}^{\infty} \frac{2^n}{n!}$? Bestimmen Sie ggf. die Summe.

💡 **Strategy:** Recall the exponential series $e^x = \sum x^n/n!$.

Solution: Converges; sum = $e^2 - 1$.

🔥 Working space

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Section 4: Mock Exam 2 (Q11–Q20)

Q11 Medium Thermodynamics 6m

EN: An ideal gas at 2 atm and 300 K occupies 5 L. It expands isothermally to 15 L. Find the final pressure.

DE: Ideales Gas bei 2 atm und 300 K, 5 L. Isotherme Expansion auf 15 L. Enddruck?

Solution: $P_2 = 2/3 \text{ atm} \approx 0.67 \text{ atm}$.

 Working space

Q12 Hard Optics 8m

EN: Light passes from glass ($n = 1.5$) to air. Find the critical angle for total internal reflection.

DE: Licht von Glas ($n = 1.5$) in Luft. Kritischer Winkel?

Solution: $\sin \theta_c = 1/1.5 \Rightarrow \theta_c \approx 41.8^\circ$.

 Working space

Q13 Hard Atomic 8m

EN: Calculate the energy of a photon with wavelength 500 nm in eV. ($h = 6.63 \cdot 10^{-34} \text{ J}\cdot\text{s}$, $c = 3 \cdot 10^8 \text{ m/s}$, $1 \text{ eV} = 1.6 \cdot 10^{-19} \text{ J}$)

DE: Energie eines Photons mit $\lambda = 500 \text{ nm}$ in eV.

Solution: $E \approx 2.49 \text{ eV}$.

 Working space

Q14 **Hard** Chemistry 7m

EN: Balance the following redox reaction in acidic solution: $\text{MnO}_4^- + \text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+}$.

DE: Redox-Gleichung in saurer Lösung ausgleichen.

Solution: $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$.

 Working space

Q15 **Medium** Chemistry 6m

EN: Calculate the molarity of a solution containing 5.85 g of NaCl in 500 mL of solution. ($M_{\text{NaCl}} = 58.5$ g/mol)

DE: Molarität: 5.85 g NaCl in 500 mL.

Solution: 0.2 M.

 Working space

Q16 **Medium** Biology 6m

EN: Explain the role of ATP in cellular processes and why it is called the "energy currency" of the cell.

DE: Rolle von ATP in Zellprozessen.

Solution: ATP stores and transfers energy; hydrolysis to ADP + Pi releases usable energy for cellular work.

 Working space

Q17 **Hard** Biology 8m

EN: A cross between two heterozygous tall plants (Tt) produces 400 offspring. Predict how many will be short. (T = tall dominant, t = short recessive)

DE: Kreuzung zweier heterozygoter großer Pflanzen (Tt), 400 Nachkommen. Wie viele klein?

Solution: 25% short → 100 short offspring.

 Working space

Q18 Medium Physics 6m

EN: A wave has frequency 200 Hz and wavelength 1.5 m. What is the period and speed?

DE: Welle mit 200 Hz und 1.5 m. Periode und Geschwindigkeit?

Solution: $T = 0.005 \text{ s}$; $v = 300 \text{ m/s}$. Working space

Q19 Hard Chemistry 8m

EN: Explain the difference between an endothermic and an exothermic reaction. Give one example of each.

DE: Unterschied zwischen endothermer und exothermer Reaktion. Je ein Beispiel.

Solution: Exothermic releases heat (combustion); endothermic absorbs heat (photosynthesis, melting ice). Working space

Q20 Hard Mixed 10m

EN: In a population of 1000 individuals, 160 show a recessive trait. Using Hardy–Weinberg, find the allele frequencies p and q , and the number of heterozygous carriers.DE: 1000 Individuen, 160 mit rezessivem Merkmal. Hardy–Weinberg: p , q , Anzahl heterozygoter Träger. **Strategy:** $q^2 = 160/1000 = 0.16$, $q = 0.4$, $p = 0.6$, $2pq = 0.48$.**Solution:** $q = 0.4$, $p = 0.6$, heterozygotes = 480. Working space

Section 5: Mock Exam 3 (Q21–Q30)

Q21 **Hard** Calculus 10m

EN: Find the area between the curves $y = x^2$ and $y = x + 2$.

DE: Fläche zwischen $y = x^2$ und $y = x + 2$.

Solution: Intersections at $x = -1, 2$. Area = $\int_{-1}^2 (x + 2 - x^2) dx = 4.5$.

Working space

Q22 **Medium** Probability 6m

EN: A bag contains 5 red, 3 blue, and 2 green balls. Two balls are drawn without replacement. Find the probability both are red.

DE: 5 rot, 3 blau, 2 grün. Zwei Kugeln ohne Zurücklegen. Wahrscheinlichkeit beide rot?

Solution: $\frac{5}{10} \cdot \frac{4}{9} = \frac{2}{9} \approx 0.222$.

Working space

Q23 **Hard** Probability 8m

EN: A test for a disease has 95% sensitivity and 90% specificity. If prevalence is 2%, find $P(\text{disease}|\text{positive})$.

DE: Test mit 95% Sensitivität, 90% Spezifität. Prävalenz 2%. $P(\text{krank}|\text{positiv})$.

Solution: $P(+) = 0.019 + 0.098 = 0.117$; $P(D|+) \approx 0.162$ (%16.2).

Working space

Q24 Medium Linear Algebra 7m

EN: Solve the system: $2x + 3y = 7$, $x - y = 1$.

DE: Lösen Sie das Gleichungssystem.

Solution: $x = 2$, $y = 1$. Working space

Q25 Schwer DE Grammatik 6m

Ergänzen Sie das richtige Verb im Konjunktiv II:

„Wenn er Zeit ____, ____ er uns helfen.“

A) hätte / würde B) hat / wird C) hatte / wurde D) hätte / werde

Lösung: A) hätte / würde. Working space

Q26 Schwer DE Leseverstehen 10m

Lesen Sie den Text und antworten Sie:

„Die Globalisierung hat den internationalen Handel gefördert, aber auch zu wachsender Ungleichheit geführt. Während einige Länder von günstigeren Produkten und neuen Märkten profitieren, leiden andere unter Arbeitsplatzverlusten und Umweltbelastungen.“

Frage: Welche zwei entgegengesetzten Wirkungen der Globalisierung nennt der Text?**Lösung:** Förderung des Handels vs. wachsende Ungleichheit und Umweltbelastung. Working space

Q27 **Schwer** DE Erörterung 20m

Schreiben Sie eine Erörterung (ca. 200 Wörter): „*Sollten Universitäten kostenlos sein?*“ Einleitung, Pro, Contra, Schluss mit eigener Meinung.

Hinweis: Freies Schreiben.

 Working space

Q28 **Medium** Statistics 6m

EN: The heights of 500 students are normally distributed with $\mu = 170$ cm and $\sigma = 8$ cm. Approximately how many students are taller than 182 cm?

DE: 500 Studenten, $\mu = 170$, $\sigma = 8$. Wie viele größer als 182 cm?

 **Strategy:** $z = (182 - 170)/8 = 1.5$, $P(Z > 1.5) \approx 0.0668$.

Solution: ≈ 33 students.

 Working space

Q29 **Hard** Optimization 10m

EN: A cylindrical can must hold 500 mL. Find the radius and height that minimize the material used.

DE: Zylindrische Dose, 500 mL. Minimieren Sie das Material.

Solution: $h = 2r$, $r \approx 4.3$ cm, $h \approx 8.6$ cm. Working spaceQ30 **Hard** Calculus 10mEN: Evaluate $\int_0^\pi \sin^2(x) dx$.DE: Berechnen Sie $\int_0^\pi \sin^2(x) dx$. **Strategy:** Use $\sin^2(x) = \frac{1 - \cos(2x)}{2}$.**Solution:** $\frac{\pi}{2}$. Working space

Section 6: Final Mixed Challenge (Q31–Q40)

The final stretch. Combine everything you have learned.

Q31 **Hard** **Mixed** **10m**

EN: Solve the differential equation $\frac{dy}{dx} = y \cos(x)$ with $y(0) = 2$.

DE: Lösen Sie $\frac{dy}{dx} = y \cos(x)$ mit $y(0) = 2$.

Solution: $y = 2e^{\sin(x)}$.

 Working space

Q32 **Hard** **Mixed** **10m**

EN: A projectile is launched at 25 m/s at 60° . Find its range on horizontal ground ($g = 10 \text{ m/s}^2$).

DE: Projektil mit 25 m/s unter 60° . Reichweite ($g = 10 \text{ m/s}^2$).

Solution: Range = $\frac{v^2 \sin(2\theta)}{g} = \frac{625 \cdot 0.866}{10} \approx 54.1 \text{ m}$.

 Working space

Q33 **Hard** **Mixed** **12m**

EN: Given $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$, find A^{-1} and verify $A \cdot A^{-1} = I$.

DE: Für $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$: A^{-1} berechnen und prüfen.

Solution: $A^{-1} = \frac{1}{10} \begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix}$.

 Working space

Q34 Hard Mixed 10m

EN: Compute the sum $\sum_{k=0}^5 \binom{5}{k}$ using the binomial theorem.DE: Summe $\sum_{k=0}^5 \binom{5}{k}$ mit dem Binomialsatz.**Solution:** $= 2^5 = 32$.

Working space

Q35 Schwer DE Textanalyse 10m

Analysieren Sie den Satz: „Obwohl die Technologie fortschreitet, bleibt der Mensch die zentrale Figur.“
 Erklären Sie Konjunktion, Wortstellung und Bedeutung.

Lösung: „Obwohl“ leitet Nebensatz ein (Verb am Ende); Hauptsatz danach beginnt mit Verb an 2. Stelle.
 Bedeutung: Kontrast zwischen Fortschritt und menschlicher Rolle.

Working space

Q36 Hard Mixed 12m

EN: A 2 kg block is attached to a spring ($k = 200$ N/m) and oscillates with amplitude 0.1 m. Find the total energy and maximum speed.DE: 2 kg Block an Feder $k = 200$ N/m, Amplitude 0.1 m. Gesamtenergie und maximale Geschwindigkeit.**Solution:** $E = \frac{1}{2}kA^2 = 1$ J; $v_{max} = \sqrt{k/m} \cdot A = 1$ m/s.

Working space

Q37 Hard Mixed 10m

EN: Use L'Hôpital to evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.DE: L'Hôpital: $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.**Solution:** $\frac{1}{2}$.

Working space

Q38 Hard Mixed 10m

EN: Show that $\cos(2x) = 1 - 2 \sin^2(x)$ using double-angle identities.DE: Zeigen Sie $\cos(2x) = 1 - 2 \sin^2(x)$.**Solution:** $\cos(2x) = \cos^2 x - \sin^2 x = 1 - \sin^2 x - \sin^2 x = 1 - 2 \sin^2 x$. Working space

Q39 Schwer DE Aufsatz 20m

Schreiben Sie einen Aufsatz (ca. 250 Wörter) zum Thema: „*Welche Rolle spielt Mathematik in der modernen Welt?*“ Einleitung, Argumente, Beispiele, Schluss.**Hinweis:** Freies Schreiben. Working space

Q40 Hard Final Reflection —

EN: Write a short paragraph (5–6 sentences) in English: What did you learn from this 5-year preparation journey? What are your next steps?

DE: Kurzer Absatz auf Deutsch: Was hast du aus dieser 5-jährigen Vorbereitung gelernt? Was sind deine nächsten Schritte?

Hinweis: This is for you. There is no right or wrong answer.

 Working space

Answer Key (Q1 – Q40)

Q	Answer	Q	Answer	Q	Answer	Q	Answer
Q1	$-\tan x + e^{3x}(3 \sin x + \cos x)$	Q11	$\approx 0.67 \text{ atm}$	Q21	4.5	Q31	$y = 2e^{\sin x}$
Q2	$\ln 2$	Q12	$\approx 41.8^\circ$	Q22	$2/9 \approx 0.222$	Q32	$\approx 54.1 \text{ m}$
Q3	$(-10, 7, 9);$ $\cos \theta = -8/(\sqrt{14}\sqrt{21})$	Q13	$\approx 2.49 \text{ eV}$	Q23	$\approx 16.2\%$	Q33	$\frac{1}{10} \begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix}$
Q4	$h = 11.25 \text{ m}; t = 3 \text{ s}$	Q14	Balanced redox equation	Q24	$x = 2, y = 1$	Q34	32
Q5	$\approx 4.8 \text{ m/s}$	Q15	0.2 M	Q25	A) hätte / würde	Q35	Nebensatz + Hauptsatz
Q6	$I = 6 \text{ A}, P_1 = 24 \text{ W}$	Q16	ATP energy currency	Q26	Handel vs. Ungleichheit	Q36	$E = 1 \text{ J}, v_{\max} = 1 \text{ m/s}$
Q7	$x = 4, h = 2; SA = 48 \text{ m}^2$	Q17	100 short	Q27	Free writing	Q37	1/2
Q8	$F = 0.6 \text{ N}$	Q18	$T = 0.005 \text{ s};$ $v = 300 \text{ m/s}$	Q28	$\approx 33 \text{ students}$	Q38	Identity shown
Q9	-4	Q19	Exo vs endo	Q29	$r \approx 4.3 \text{ cm},$ $h \approx 8.6 \text{ cm}$	Q39	Free writing
Q10	$e^2 - 1$	Q20	$q = 0.4, p = 0.6;$ 480	Q30	$\pi/2$	Q40	Reflection 9

Final Words

You have reached the end of the series. Over 160 questions, 4 parts, and countless hours of preparation — this is the moment where knowledge meets discipline.

The ETH Zürich entrance exam is not just a test of intelligence. It is a test of **consistency, calm under pressure, and the ability to think clearly when it matters most**. You have built the foundation. Now trust it.

Remember: every problem you solved, every mistake you logged, every late night spent on a difficult integral — all of it was preparation for this. Walk into the exam hall with confidence. You have earned it.

"The best preparation for tomorrow is doing your best today."

— H. Jackson Brown Jr.

Good luck. You've got this. ❤️

References & Copyright Notice

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- **References:** ETH Aufnahmeprüfung archive, GitHub *janikschuettler/eth-entrance-exam*, Halliday–Resnick, Tipler–Mosca, Papula, Stewart, Campbell *Biology*, Atkins *Chemistry*, Goethe-Institut materials.
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~ END OF SERIES ~